CAPSTONE PROJECT

Machine Learning in Practice

Asian Institute of Management

Credit Card Fraud Detection

A Machine Learning Approach to Financial Transaction Security

Step 2: Data Collection & Understanding

Finance Domain | Binary Classification | 2026

1. Dataset Source & Acquisition

The dataset used in this project is the Credit Card Fraud Detection dataset published by the Machine Learning Group of Université Libre de Bruxelles (ULB) and made publicly available on Kaggle. It is one of the most widely cited benchmark datasets for fraud detection research and has been used extensively in academic literature and industry benchmarking.

Attribute	Details
Dataset Name	Credit Card Fraud Detection
Source	Kaggle / ULB Machine Learning Group
URL	kaggle.com/datasets/mlg-ulb/creditcardfraud
File Format	CSV (creditcard.csv)
File Size	Approximately 144 MB
License	Open Database License (ODbL)
Data Period	September 2013 (2 days of transactions)
Geography	European cardholders
Last Updated	2018 (stable benchmark dataset)

2. Dataset Overview

The dataset contains transactions made over two days in September 2013 by European credit cardholders. The figures below reflect the published dataset as documented by ULB on Kaggle, prior to any cleaning or preprocessing steps, which are covered in Step 3.

Property	Value
Total Transactions	284,807
Fraudulent Transactions	492 (0.173% — as published by ULB)
Legitimate Transactions	284,315 (99.827%)
Total Features	31 (30 input features + 1 target)
Missing Values	None documented by ULB
Target Variable	Class (0 = Legitimate, 1 = Fraud)
Time Span	48 hours (172,792 seconds)

3. Feature Summary

The dataset contains three categories of features: PCA-transformed anonymized features, two interpretable numerical features, and the binary target variable.

Category	Features	Count	Type
PCA-Transformed	V1, V2, V3, ... V28	28	Continuous (Float)
Interpretable	Time, Amount	2	Continuous (Float)
Target Variable	Class	1	Binary Integer (0 or 1)

4. Data Dictionary

4.1 PCA-Transformed Features (V1 – V28)

Feature	Type	Range	Description
V1	Float	Unbounded	1st principal component of the original transaction features
V2	Float	Unbounded	2nd principal component of the original transaction features
V3	Float	Unbounded	3rd principal component of the original transaction features
V4	Float	Unbounded	4th principal component of the original transaction features
V5	Float	Unbounded	5th principal component of the original transaction features
V6	Float	Unbounded	6th principal component of the original transaction features
V7	Float	Unbounded	7th principal component of the original transaction features
V8	Float	Unbounded	8th principal component of the original transaction features
V9	Float	Unbounded	9th principal component of the original transaction features
V10	Float	Unbounded	10th principal component of the original transaction features
V11	Float	Unbounded	11th principal component of the original transaction features
V12	Float	Unbounded	12th principal component of the original transaction features
V13	Float	Unbounded	13th principal component of the original transaction features
V14	Float	Unbounded	14th principal component of the original transaction features
V15	Float	Unbounded	15th principal component of the original transaction features
V16	Float	Unbounded	16th principal component of the original transaction features
V17	Float	Unbounded	17th principal component of the original transaction features
V18	Float	Unbounded	18th principal component of the original transaction features
V19	Float	Unbounded	19th principal component of the original transaction features
V20	Float	Unbounded	20th principal component of the original transaction features
V21	Float	Unbounded	21st principal component of the original transaction features
V22	Float	Unbounded	22nd principal component of the original transaction features
V23	Float	Unbounded	23rd principal component of the original transaction features
V24	Float	Unbounded	24th principal component of the original transaction features
V25	Float	Unbounded	25th principal component of the original transaction features
V26	Float	Unbounded	26th principal component of the original transaction features
V27	Float	Unbounded	27th principal component of the original transaction features
V28	Float	Unbounded	28th principal component of the original transaction features

4.2 Interpretable Features

Feature	Type	Range	Unit	Description
Time	Float	0 – 172,792	Seconds	Elapsed seconds between each transaction and the first transaction in the dataset. Useful for detecting time-of-day fraud patterns.
Amount	Float	0 – 25,691.16	EUR (€)	The monetary value of the transaction. Higher-value transactions may carry different fraud risk profiles.

4.3 Target Variable

Feature	Type	Allowed Values	Description
Class	Integer	0 or 1	Binary fraud label. 0 = legitimate transaction; 1 = confirmed fraudulent transaction. Ground truth was established through manual investigation by ULB researchers.

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5. Initial Observations & Motivation

5.1 On Class Imbalance

The target variable is extremely unbalanced and this is not unusual. In fact, it is expected.

Fraudulent transactions are rare precisely because financial institutions already employ safeguards to prevent them from occurring. Whether those safeguards are traditional rule-based systems or more recent machine learning approaches, their existence means that by the time a transaction dataset is collected, the vast majority of fraud has already been stopped. What remains in the data is the fraud that slipped through — a small but critical subset.

This imbalance is not a data quality problem; it is a reflection of reality, and any model built on this data must be designed with this reality in mind. Standard accuracy is therefore not a meaningful evaluation metric for this project — a model that predicts every transaction as legitimate would achieve over 99% accuracy while detecting zero fraud. The primary metrics used throughout this project are AUC-ROC, F1-Score, and Precision-Recall AUC, all of which are robust to class imbalance.

5.2 On PCA-Transformed Features

The input features V1 through V28 have been processed through Principal Component Analysis (PCA) by the original ULB researchers. This was done deliberately to protect the identity and personal data of the cardholders in the dataset. PCA abstracts the original transaction details into mathematical components, making it impossible to trace back to any individual.

While this means we cannot directly interpret what each V feature represents in business terms, the components still retain the statistical patterns present in the original data — they remain relevant and usable for building a predictive model. What is lost is direct interpretability; what is preserved is predictive signal. SHAP (SHapley Additive exPlanations) analysis in Step 3 will be used to explain model decisions in terms of these transformed features, providing a layer of explainability even without access to the original variables.

5.3 Why This Dataset

I chose this dataset because credit card fraud represents a real, continuously evolving problem with no permanent solution.

Unlike academic exercises built around clean, stable problems, fraud detection is a living challenge, fraudsters adapt, patterns shift, and yesterday's detection method becomes tomorrow's blind spot. Beyond the technical challenge, the stakes are deeply human. Credit cards and digital payments are now part of daily life for hundreds of millions of people, and the financial and psychological impact of fraud on individuals is real.

Finding better ways to detect it even incrementally has direct relevance to a large portion of the population. That practical significance is what drew me to this domain. Furthermore, while this dataset reflects European cardholder behavior, the methodology developed here is designed to be transferable. Directly applicable to the Philippine digital payments context once local transaction data becomes available.
